Digital SAT Math

Topic 1: Quadratic Equations - Solving

Practice Questions

Total: 60 Questions

Format: Digital SAT Style — MCQ & Grid-In

Difficulty: Easy (Q1-20) • Medium (Q21-45) • Hard (Q46-60)

Q1.

DIFFICULTY: Easy

Which of the following is a solution to the equation $x^2 - 9x + 14 = 0$?

A) -7
B) -2
C) 2
D) 9

Q2.

DIFFICULTY: Easy

$x^2 - 16 = 0$

What is the positive solution to the given equation?

[GRID-IN — Enter your answer below]

Q3.

DIFFICULTY: Easy

Which of the following is equivalent to $x^2 + 10x + 25$?

A) $(x + 5)^2$
B) $(x - 5)^2$
C) $(x + 5)(x - 5)$
D) $(x + 25)(x + 1)$

Q4.

DIFFICULTY: Easy

The equation $x^2 - 6x = 0$ has two solutions. What is the positive solution?

[GRID-IN — Enter your answer below]

Q5.

DIFFICULTY: Easy

Which of the following equations has no real solutions?

A) $x^2 - 4 = 0$
B) $x^2 + 4 = 0$
C) $x^2 - 4x = 0$
D) $x^2 + 4x - 5 = 0$

Q6.

DIFFICULTY: Easy

A rectangle has a length that is 3 more than its width. If the area of the rectangle is 40 square centimeters, which of the following equations represents this situation, where w is the width, in centimeters?

A) $w^2 + 3 = 40$
B) $2w + 3 = 40$
C) $w(w + 3) = 40$
D) $w + (w + 3) = 40$

Q7.

DIFFICULTY: Easy

$2x^2 - 8 = 0$

What is the positive value of x?

[GRID-IN — Enter your answer below]

Q8.

DIFFICULTY: Easy

Which of the following is a factor of $x^2 - 7x + 12$?

A) $(x - 2)$
B) $(x - 3)$
C) $(x + 4)$
D) $(x + 6)$

Q9.

DIFFICULTY: Easy

For the quadratic equation $x^2 + bx + 18 = 0$, one solution is $x = 2$. What is the value of b?

A) -11
B) -9
C) 9
D) 11

Q10.

DIFFICULTY: Easy

Which of the following is equivalent to $(x + 7)(x - 3)$?

A) $x^2 + 4x - 21$
B) $x^2 - 4x - 21$
C) $x^2 + 4x + 21$
D) $x^2 - 10x + 21$

Q11.

DIFFICULTY: Easy

The graph of $y = x^2 - 4x + 3$ in the xy-plane has x-intercepts at two points. What is the sum of the x-coordinates of these two points?

A) -4
B) 1
C) 3
D) 4

Q12.

DIFFICULTY: Easy

$x^2 + 12x + 36 = 0$

What is the value of x?

[GRID-IN — Enter your answer below]

Q13.

DIFFICULTY: Easy

Which of the following equations has exactly one real solution?

A) $x^2 - 9 = 0$
B) $x^2 + 9 = 0$
C) $x^2 - 6x + 9 = 0$
D) $x^2 - 6x + 8 = 0$

Q14.

DIFFICULTY: Easy

$3x^2 = 75$

What is the positive value of x?

[GRID-IN — Enter your answer below]

Q15.

DIFFICULTY: Easy

The function f is defined by $f(x) = x^2 - 5x + 6$. If $x > 2$, what is the value of x for which $f(x) = 0$?

A) 2
B) 3
C) 5
D) 6

Q16.

DIFFICULTY: Easy

Which of the following is a solution to $(x - 4)(x + 9) = 0$?

A) -9
B) -4
C) 5
D) 13

Q17.

DIFFICULTY: Easy

A ball is launched upward. The height h, in feet, of the ball t seconds after launch is given by $h = -16t^2 + 32t$. For what positive value of t does the ball return to ground level?

A) 1
B) 2
C) 4
D) 16

Q18.

DIFFICULTY: Easy

Which of the following correctly factors $x^2 - 49$?

A) $(x - 7)^2$
B) $(x + 7)^2$
C) $(x + 7)(x - 7)$
D) $(x - 49)(x + 1)$

Q19.

DIFFICULTY: Easy

The solutions to a quadratic equation are $x = 3$ and $x = -5$. Which of the following could be the equation?

A) $x^2 + 2x - 15 = 0$
B) $x^2 - 2x - 15 = 0$
C) $x^2 + 8x + 15 = 0$
D) $x^2 - 8x + 15 = 0$

Q20.

DIFFICULTY: Easy

$x^2 - 11x + 28 = 0$

If $x > 5$, what is the value of x?

[GRID-IN — Enter your answer below]

Q21.

DIFFICULTY: Medium

$(2x - 1)^2 - 9(2x - 1) + 20 = 0$

What is the smaller solution to the given equation?

A) 3/2
B) 5/2
C) 3
D) 7/2

Q22.

DIFFICULTY: Medium

In the xy-plane, the graph of $y = x^2 + kx - 12$ passes through the point (-3, 0), where k is a constant. What is the value of k?

A) -7
B) -1
C) 1
D) 7

Q23.

DIFFICULTY: Medium

The equation $4x^2 - 12x + 9 = 0$ has one real solution. What is that solution?

A) 1/2
B) 3/4
C) 3/2
D) 3

Q24.

DIFFICULTY: Medium

The height h, in meters, of an object t seconds after being launched vertically is modeled by $h = -5t^2 + 30t + 8$. What is the maximum height, in meters, reached by the object?

A) 38
B) 53
C) 83
D) 133

Q25.

DIFFICULTY: Medium

Which of the following is an equivalent form of $6x^2 + 7x - 3$?

A) (2x + 3)(3x - 1)
B) (6x - 1)(x + 3)
C) (2x - 1)(3x + 3)
D) (3x + 1)(2x - 3)

Q26.

DIFFICULTY: Medium

$y = x^2 - 4x + c$

In the given equation, c is a constant. The equation has two distinct real solutions. Which of the following could be the value of c?

A) 3
B) 4
C) 5
D) 6

Q27.

DIFFICULTY: Medium

A company's monthly profit P, in thousands of dollars, is modeled by $P = -x^2 + 14x - 40$, where x is the number of units produced per day. For what values of x is the profit equal to zero?

A) $x = 2$ and $x = 20$
B) $x = 4$ and $x = 10$
C) $x = 5$ and $x = 8$
D) $x = 7$ and $x = 7$

Q28.

DIFFICULTY: Medium

$x^2 + 8x + 7 = -8$

What is the sum of all solutions to the given equation?

A) -15
B) -8
C) 8
D) 15

Q29.

DIFFICULTY: Medium

The quadratic function f is defined by $f(x) = -2(x - 3)^2 + 18$. What is the x-coordinate of the vertex of the parabola $y = f(x)$?

[GRID-IN — Enter your answer below]

Q30.

DIFFICULTY: Medium

If $(x - p)(x - q) = x^2 - 5x + 6$, where p and q are constants with $p > q$, what is the value of $p - q$?

[GRID-IN — Enter your answer below]

Q31.

DIFFICULTY: Medium

The function g is defined by $g(x) = x^2 - 6x + 5$. For how many values of x does $g(x) = -3$?

A) 0
B) 1
C) 2
D) 3

Q32.

DIFFICULTY: Medium

A farmer encloses a rectangular garden using 60 feet of fencing on three sides, with a barn wall as the fourth side. The side parallel to the barn has width w feet. Which equation represents an area of 448 square feet?

A) $w(60 - w) = 448$
B) $w(60 - 2w) = 448$
C) $2w + 60 = 448$
D) $w^2 + 60w = 448$

Q33.

DIFFICULTY: Medium

In the xy-plane, the parabola $y = (x - 2)(x - 8)$ has a vertex at (h, k). What is the value of k?

A) -16
B) -9
C) 5
D) 10

Q34.

DIFFICULTY: Medium

$2x^2 + 5x - 12 = 0$

If $x > 0$, what is the value of x?

A) 3/2
B) 2
C) 3
D) 4

Q35.

DIFFICULTY: Medium

The equation $x^2 + mx + n = 0$ has solutions $x = -2$ and $x = 7$. What are the values of m and n, respectively?

A) $m = -5, n = -14$
B) $m = 5, n = -14$
C) $m = -5, n = 14$
D) $m = 5, n = 14$

Q36.

DIFFICULTY: Medium

$y = x^2 - 2x - 3$

$y = 2x - 3$

The given system has two solutions. What is the sum of the x-values of these solutions?

A) 0
B) 2
C) 4
D) 6

Q37.

DIFFICULTY: Medium

$3x^2 - 27 = 0$

Which of the following is equivalent to the given equation?

A) $x^2 = 3$
B) $x^2 = 9$
C) $(x - 3)^2 = 0$
D) $x(x - 3) = 0$

Q38.

DIFFICULTY: Medium

A quadratic function f has its vertex at (4, -7). If f(0) = 9, which of the following defines f?

A) $f(x) = (x - 4)^2 - 7$
B) $f(x) = (x + 4)^2 - 7$
C) $f(x) = (x - 4)^2 + 9$
D) $f(x) = 2(x - 4)^2 - 7$

Q39.

DIFFICULTY: Medium

The function p is defined by $p(x) = x^2 - x - 6$. The function q is defined by $q(x) = p(x + 4)$. If $x < -2$, what is the value of x for which $q(x) = 0$?

A) -6
B) -5
C) -2
D) -1

Q40.

DIFFICULTY: Medium

For the equation $3x^2 - 6x + k = 0$, the discriminant is zero. What is the value of k?

[GRID-IN — Enter your answer below]

Q41.

DIFFICULTY: Medium

In the xy-plane, the graph of $y = ax^2 + bx + c$ passes through (0, 5), (1, 0), and (5, 0). What is the value of $a + b + c$?

A) -5
B) 0
C) 1
D) 5

Q42.

DIFFICULTY: Medium

Which of the following has the same x-intercepts as $y = x^2 - x - 12$?

A) $y = (x + 4)(x - 3)$
B) $y = -2(x - 4)(x + 3)$
C) $y = 3(x + 4)(x - 3)$
D) $y = -(x + 4)(x - 3)$

Q43.

DIFFICULTY: Medium

The width of a rectangular pool is 7 feet less than its length. The area is 120 square feet. What is the length, in feet?

A) 8
B) 10
C) 12
D) 15

Q44.

DIFFICULTY: Medium

$x^2 + (k + 2)x + 2k = 0$

In the given equation, k is a positive constant. Which of the following is always a solution, regardless of the value of k?

A) $x = -k$
B) $x = -2$
C) $x = k$
D) $x = 2$

Q45.

DIFFICULTY: Medium

The equation $(x - 3)^2 = 25$ has two solutions. What is the product of its solutions?

A) -16
B) -8
C) 8
D) 16

Q46.

DIFFICULTY: Hard

$f(x) = x^2 - 8x + 12$ and $g(x) = f(x + 3)$. Which of the following is equivalent to $g(x)$?

A) $x^2 - 2x - 3$
B) $x^2 + 2x + 3$
C) $x^2 - 14x + 39$
D) $x^2 - 2x + 3$

Q47.

DIFFICULTY: Hard

$y = x^2 - 10x + c$

$y = 3 - 2x$

For what value of c does the given system have exactly one real solution?

A) 15
B) 17
C) 19
D) 22

Q48.

DIFFICULTY: Hard

$kx^2 + 4x + (k - 3) = 0$, where k is a nonzero constant, has two equal real solutions. What is a possible value of k?

A) -4
B) -1
C) 1
D) 4

Q49.

DIFFICULTY: Hard

$f(x) = 2x^2 - 12x + 19$. Which form displays the minimum value of f as a constant or coefficient?

A) $f(x) = 2(x - 3)^2 + 1$
B) $f(x) = 2(x - 3)^2 + 7$
C) $f(x) = 2(x - 6)^2 + 1$
D) $f(x) = 2x(x - 6) + 19$

Q50.

DIFFICULTY: Hard

For $ax^2 + bx + c = 0$ with nonzero constants, the discriminant $b^2 - 4ac = 0$. Which statement must be true?

A) There are no real solutions.
B) There are exactly two distinct real solutions.
C) There is exactly one real solution, equal to $-b/(2a)$.
D) There are two solutions, each equal to $b/(2a)$.

Q51.

DIFFICULTY: Hard

$x^2 - 4x + 1 = 0$

One solution can be written as p + q*(sqrt 3), where p and q are integers. What is p + q?

A) 1
B) 2
C) 3
D) 4

Q52.

DIFFICULTY: Hard

Selected values of quadratic function f are shown.

x | f(x)

-1 | 6

0 | 1

1 | -2

2 | -3

3 | -2

Which of the following could define f?

A) $f(x) = (x - 2)^2 - 3$
B) $f(x) = -(x - 2)^2 - 3$
C) $f(x) = (x + 2)^2 - 3$
D) $f(x) = (x + 2)^2 + 3$

Q53.

DIFFICULTY: Hard

A ball thrown from the top of a 50-foot building has height $h = -16t^2 + 24t + 50$ feet above ground t seconds after being thrown. To the nearest tenth of a second, when does it hit the ground?

A) 2.3
B) 2.7
C) 3.1
D) 3.5

Q54.

DIFFICULTY: Hard

$x^2 + y = 7$

$x - y = -1$

For the solution where $x > 0$, what is the value of x?

[GRID-IN — Enter your answer below]

Q55.

DIFFICULTY: Hard

$f(x) = (x + 2)(x - 5) + 10$ and $g(x) = (x + a)(x + b)$ define the same function. What is $|a - b|$?

A) 1
B) 3
C) 5
D) 7

Q56.

DIFFICULTY: Hard

The graph of $y = x^2 - 6x + 5$ is shifted left 2 units and up 4 units. What is the minimum value of the new parabola?

A) -4
B) 0
C) 4
D) 8

Q57.

DIFFICULTY: Hard

$x^2 + px + q = 0$ has solutions r and s with $r > s$. If $r - s = 8$ and $r + s = 4$, what is q?

A) -12
B) -20
C) -24
D) -32

Q58.

DIFFICULTY: Hard

$h(x) = x^2 + bx + c$ passes through $(6, 0)$ and has vertex at $(4, k)$. What is b?

[GRID-IN — Enter your answer below]

Q59.

DIFFICULTY: Hard

$f(x) = x^2 - 2kx + k^2$

$g(x) = x^2 - 2kx + (k^2 - 4)$

where k is a positive constant. Which of the following correctly gives the x-intercepts of g?

A) $x = k - 2$ and $x = k + 2$
B) $x = k - 4$ and $x = k + 4$
C) $x = -k - 2$ and $x = -k + 2$
D) $x = k^2 - 4$ and $x = k^2 + 4$

Q60.

DIFFICULTY: Hard

$y = 2x^2 - 8x + 3$ intersects $y = 3 - 4x$ at two points. What is the positive x-coordinate of either intersection point?

[GRID-IN — Enter your answer below]